

Quality Control and Preprocessing :

Tools and Software:

- (a) FastQC : Check the quality of the raw reads (base scores, GC content, adapter contamination).
- (b) Trimmomatic / Cutadapt : Trimming of adapters, filtering of low quality reads.
- (c) MultiQC : Aggregate quality control reports across samples.

Python Integration:

- (a) Biopython for parsing FASTQ files
- (b) Automate QC pipelines with pandas and matplotlib to summarize FastQC outputs
- (c) Create reproducible workflows with Snakemake or Nextflow (Python-based workflow managers).

[RELEVANT : familiar with using libraries like pandas and matplotlib, the codes can be seen at <https://github.com/Aashu970>]

Read Alignment & Quantification

Tools & Software :

- (a) HISAT2 : fast and sensitive mapping of next-generation sequencing reads to a population of human genomes.
- (b) SAMtools : Convert, sort and index SAM/BAM files
- (c) FeatureCounts / HTSeq : Gene-level read counts quantification.

Python Integration:

- (a) Run HISAT2 automatically using subprocess or Snakemake.
- (b) Use pysam, the Python API for SAMtools, to parse BAM files.
- (c) Generate merged count matrices in pandas for downstream analyses.

Differential Gene Expression & Analysis

Tools & Software:

- (a) DESeq2 (R), edgeR, limma-voom : Find genes that are differentially expressed.
- (b) RStudio / Bioconductor : Environment for statistical modeling.

Python Integration:

- (a) Pre-processing of count matrices using pandas and numpy.
- (b) statsmodels for statistical tests (t-test, ANOVA, regression)
- (c) Machine learning (PCA, clustering, classification) with scikit-learn.
- (d) Use rpy2 to link to R packages to run DESeq2 from Python directly.

[RELEVANT : used ML and algorithms using scikit-learn, libraries like numpy and pandas been used. Codes can be seen at my GitHub profile]

Functional Enrichment And Pathway Analysis

Tools & Software:

- (a) DAVID, g:Profiler, Reactome, KEGG : Gene Ontology and pathway enrichment.
- (b) ClusterProfiler (R) : Additional enrichment visualization

Python Integration:

- (a) Requests to query the KEGG/Reactome APIs.
- (b) Automate enrichment workflows with gseapy (Python wrapper for GSEA)
- (c) Visualize enriched pathways with matplotlib, plotly or seaborn.

[RELEVANT : familiar with KEGG database and OMICSBOX software for GO Annotation analysis, generating API for KEGG, STRING, UniProt, BRENDA to integrate backend with frontend for PLASMO-PRED tool

<https://aashu970.github.io/PLASMOPRED/>]

Gene Co-expression and Network Analysis

Tools & Software:

- (a) WGCNA (R) : Weighted Gene Coexpression Network Analysis.
- (b) Cytoscape : Visualize and analyze networks
- (c) STRING : protein-protein interaction networks

Python Integration:

- (a) Generate correlation networks using networkx or igraph.
- (b) Export networks to Cytoscape using py2cytoscape
- (c) Use scikit-learn for cluster, centrality and modularity analysis.

[RELEVANT : used STRING database for protein-protein interaction for predicting CVD risk factors, and for interaction of *P. falciparum* proteins. Generated ROC-AUC graphs for ML algorithms using Python]